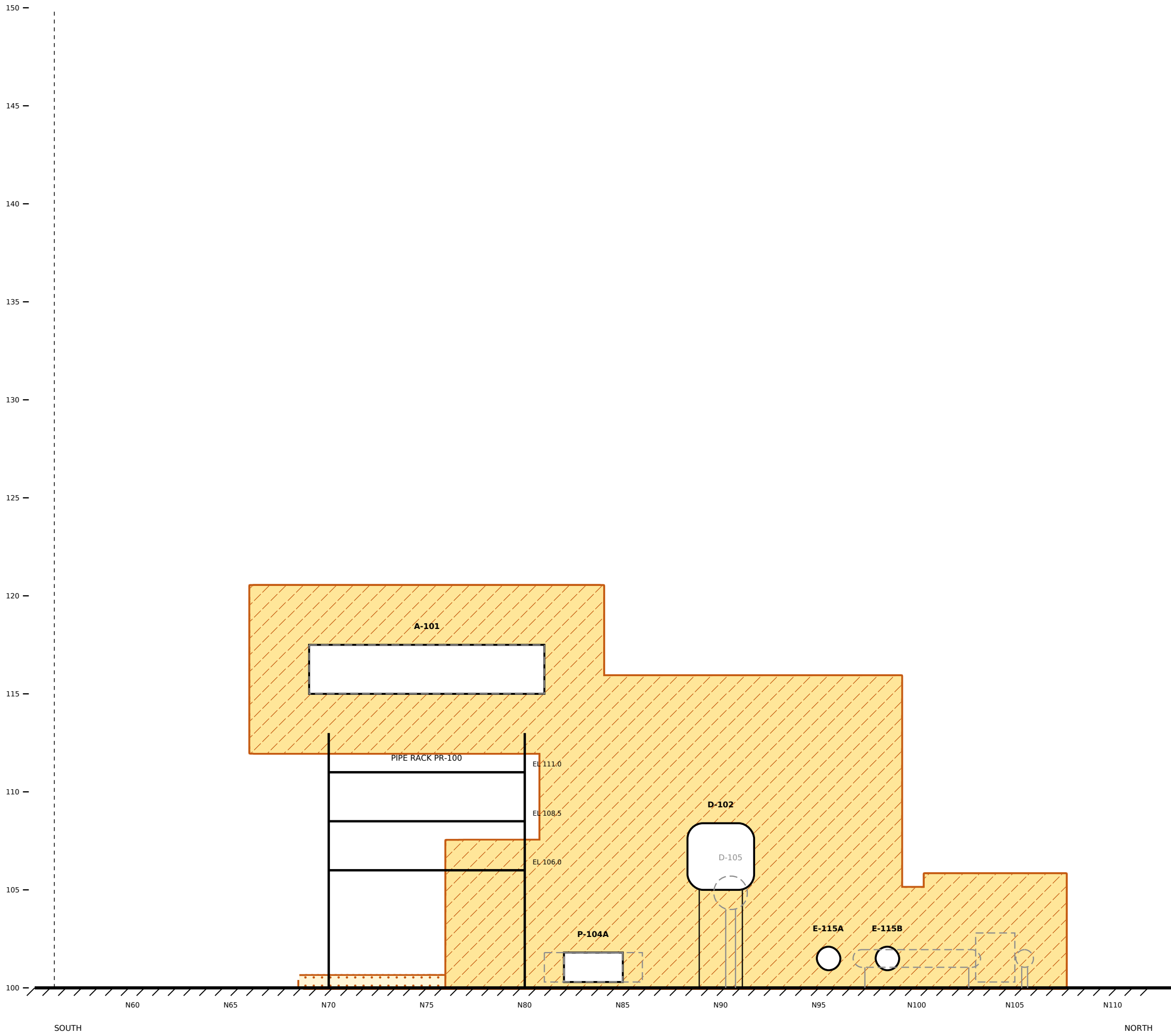


SECTION A-A (cut at E 79, looking west) SCALE 1:200

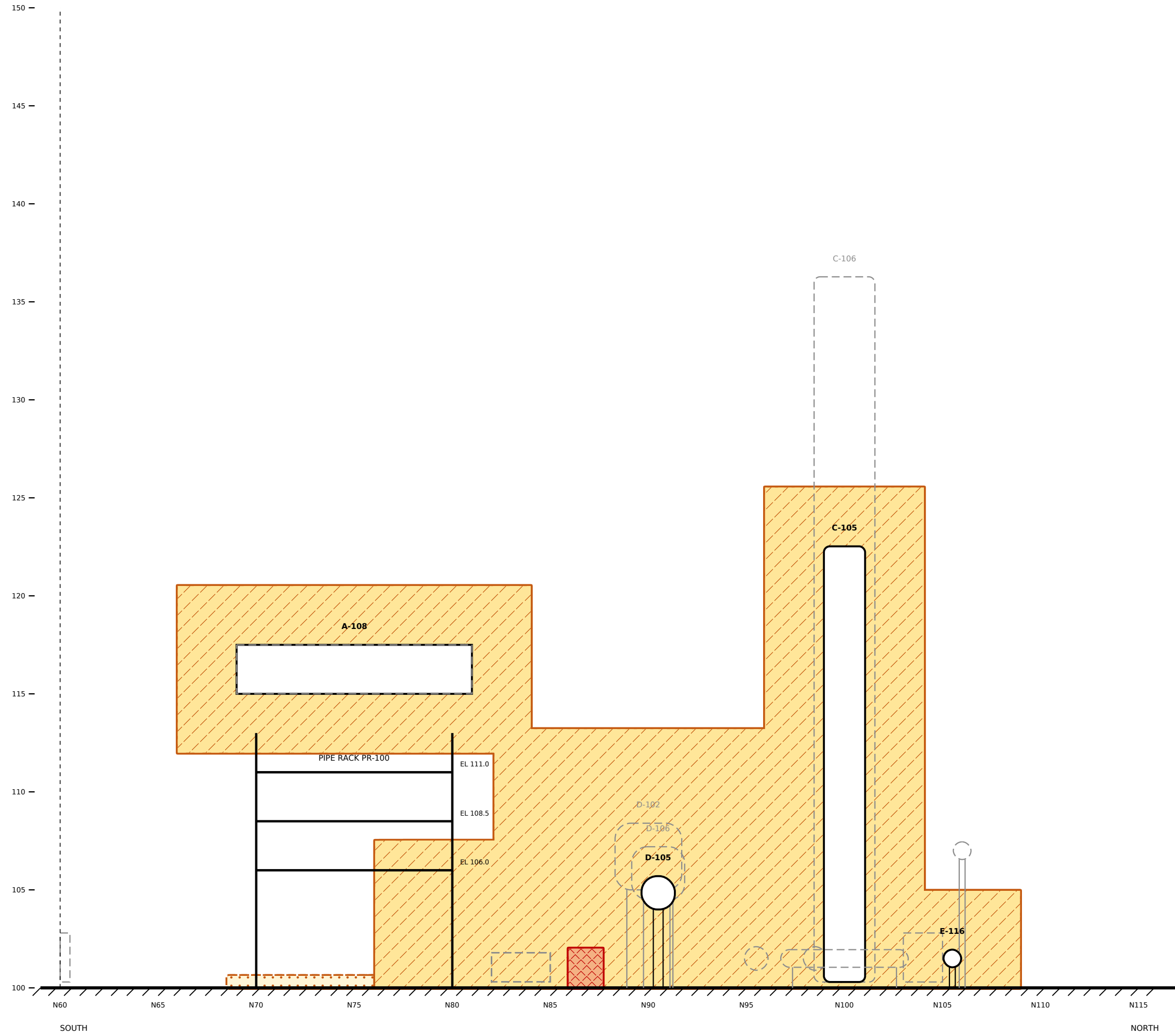


Section equipment: solid = cut, dashed grey = projected within +/- 12 m

EQUIPMENT PROTECTION BY ZONE (IEC 60079-14 / NFPA 70 Art. 505; EPL Gb / Gc)

EQUIPMENT	ZONE 1	ZONE 2
LV motors (480 V)	Ex db or Ex eb (IE, stall relay), IIB T3 (Gb)	Ex ec (non-sparking), IIA/IIB T3 (Gc); AEx ec per NEC 505
MV motors (4.16 kV)	Not located in Zone 1 (relocate) / Ex pxb	Ex ec with stator discharge risk assessment / pre-start purge, IIA T3
VFD-fed motors	Ex db certified with the drive (converter duty)	Ex ec certified for converter duty; T-class verified with VFD
Instruments	Ex ia / Ib (intrinsic safety), Ex db	Ex ia / Ic, Ex db, Ex ec
Lighting fittings	Ex db / eb, IIB T3	Ex ec / nR, IIA T3 (IIB in H25 areas)
Junction boxes / glands	Ex eb boxes; Ex db barrier glands for Ex d	Ex eb / ec boxes and glands
Heat tracing	Ex 60079-30-1, stabilised design <= T3	Ex 60079-30-1, self-regulating, T3
Desalter transformers	n/a (Zone 2 only)	Vendor package certified for Zone 2 (Ex o / ec), HV entry to vessel via Ex bushings
Welding / receptacles	Not permitted	Ex de interlocked receptacles; hot-work permit
Buildings SS-100 / FAR-100	-	Non-classified; pressurised (NFPA 496 / IEC 60079-13), intakes in unclassified area, gas detection

SECTION B-B (cut at E 64, looking west) SCALE 1:200



- NOTES**
- Sections are cut at the section lines shown on CFU-000-EL-HAC-001 (looking west). Equipment within +/- 12 m of the cut is projected (light outline); zones are shown only where the cut plane intersects them.
 - Vertical extents (heavier-than-air, open area): pump seals Zone 2 to 7.5 m above grade within 7.5 m radius; vessels / columns / exchangers Zone 2 3 m beyond the equipment, from grade to 3 m above the top source; air-cooler headers 3 m around the bundles; sample points Zone 1 1 m / Zone 2 3 m; cat. A additional Zone 2 15 m radius x 0.6 m above grade.
 - Pipe-rack tiers (EL 106.0 / 108.5 / 111.0) are unclassified except within 3 m of flanges / valves and where inside the pump / equipment envelopes shown.
 - Scale 1:200 (A1). Elevations in m, grade EL 100.000.

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
HAZARDOUS AREA CLASSIFICATION SECTIONS A-A (D-102 / OH PUMP ROW) & B-B (LPG AREA)			
DISCIPLINE	STAGE	SCALE	SHEET
ELECTRICAL	FEED	1:200	1 OF 1
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-000-EL-HAC-002			A